Machine Learning for DNA Artifact Classification
Low Template DNA (LTDNA) Analysis

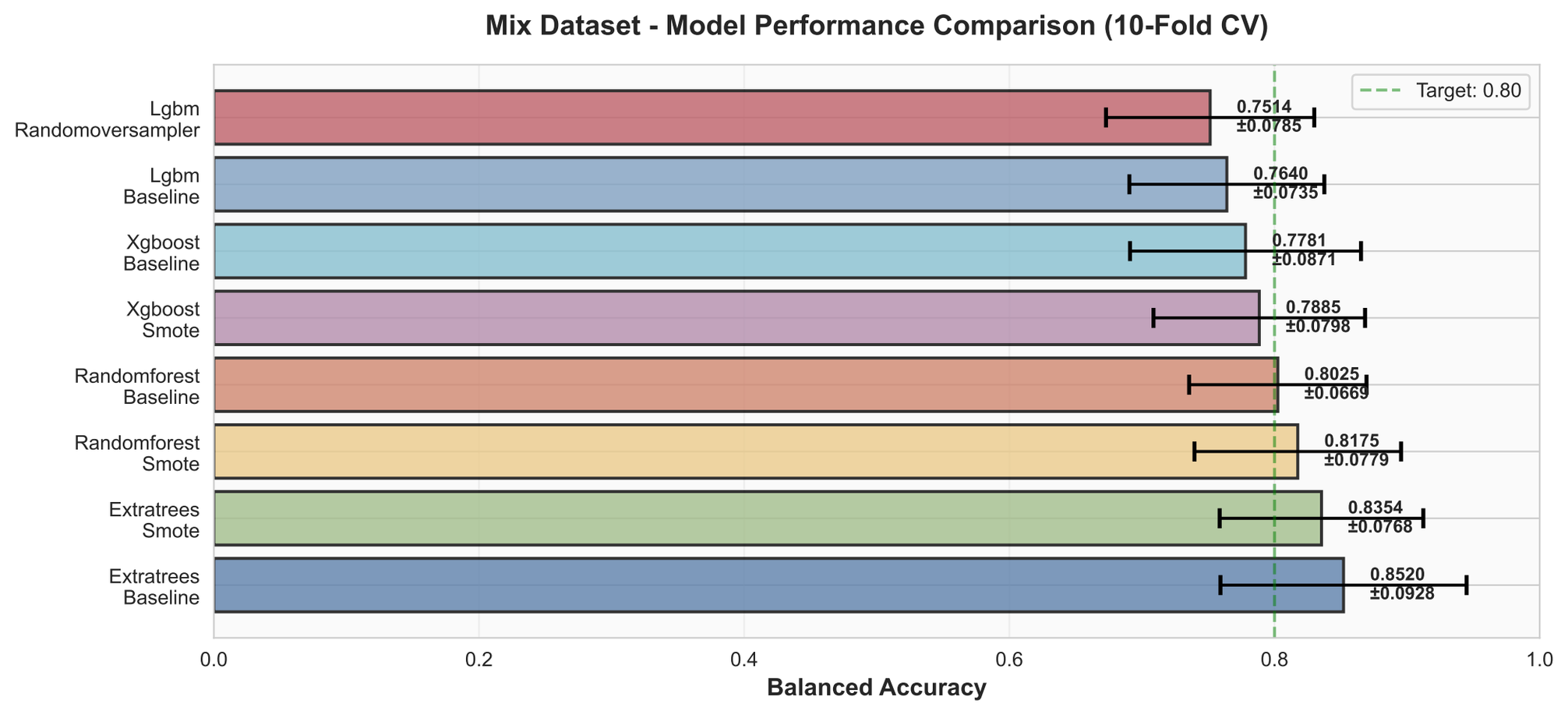
EXECUTIVE SUMMARY REPORT
Cross-Validation Results (10-Fold)

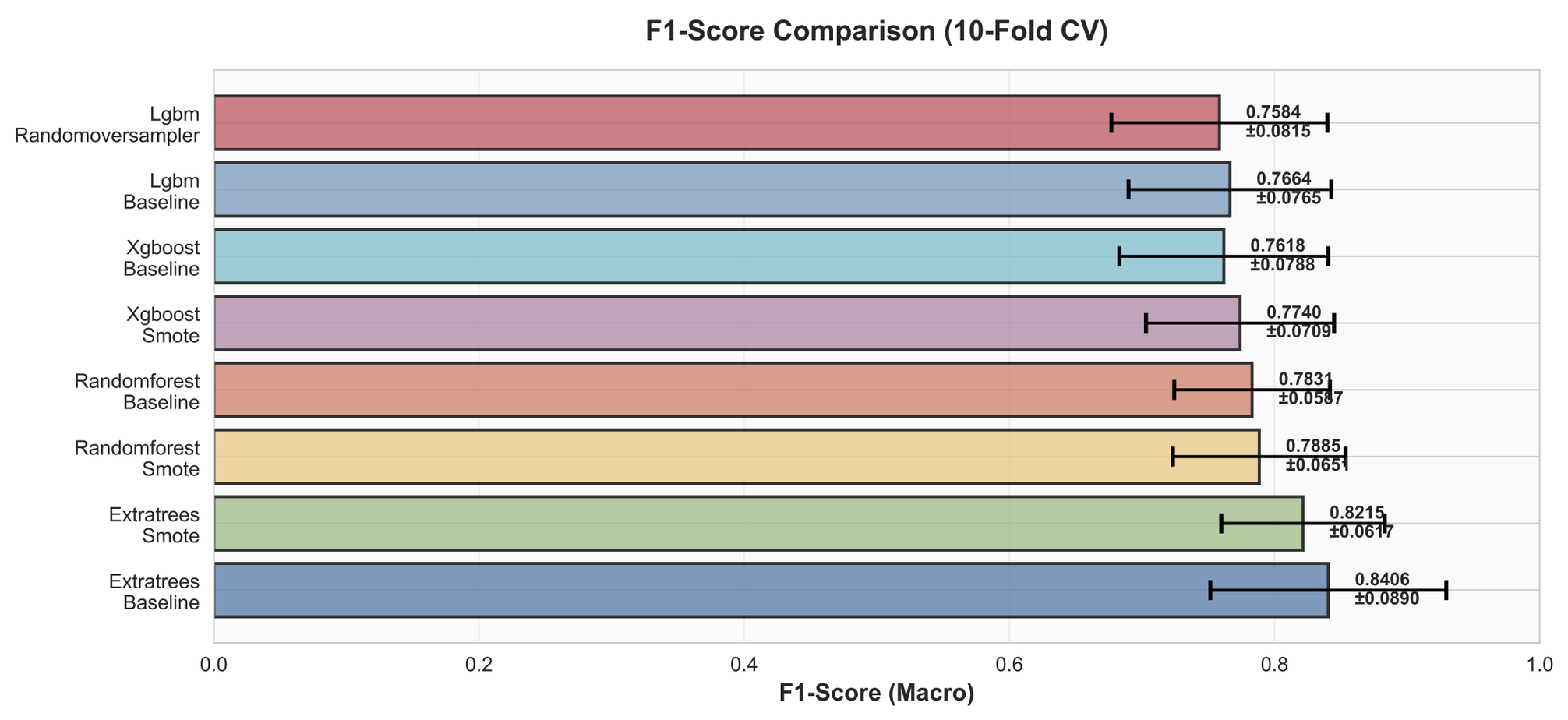
Dataset: Mix Profiles
Date: December 24, 2025
Validation: 10-Fold Stratified Cross-Validation
Models Evaluated: 8

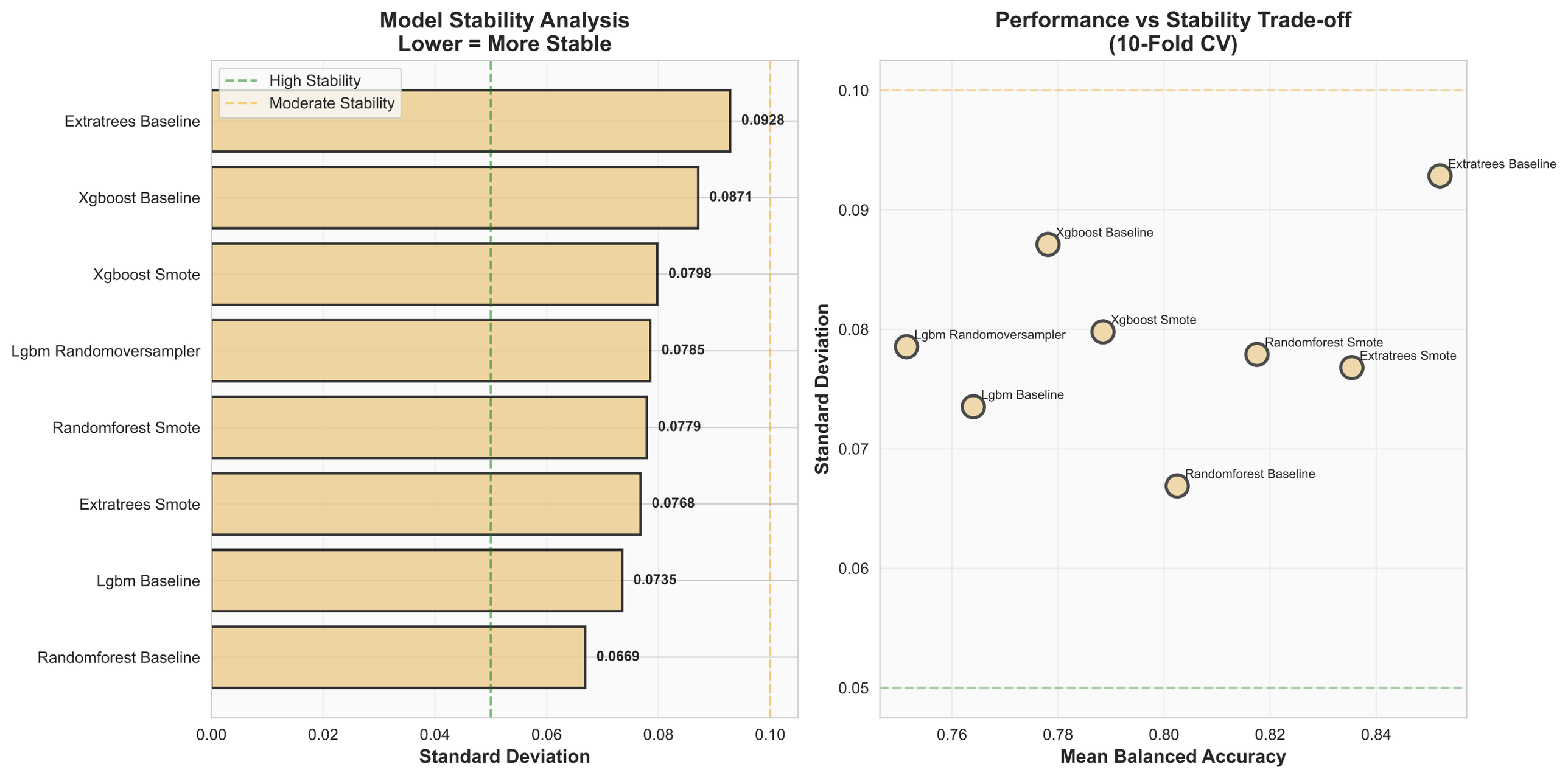
KEY FINDINGS

 Best Model: Extratrees Baseline
• Balanced Accuracy: 0.8520 ± 0.0928
• F1-Score (Macro): 0.8406
• Stability: Moderate

 Validation Approach:
• 10-fold stratified cross-validation
• Ensures robust performance estimates
• Accounts for data variability
• Reduces overfitting bias


Mix Dataset - Model Performance Comparison (10-Fold CV)
Target: 0.80
Lgbm Randomoversampler
0.7514 ±0.0785
Lgbm Baseline
0.7640 ±0.0735
Xgboost Baseline
0.7781 ±0.0871
Xgboost Smote
0.7885 ±0.0798
Randomforest Baseline
0.8025 ±0.0669
Randomforest Smote
0.8175 ±0.0779
Extratrees Smote
0.8354 ±0.0768
Extratrees Baseline
0.8520 ±0.0928
0.0
0.2
0.4
0.6
0.8
1.0
Balanced Accuracy


F1-Score Comparison (10-Fold CV)
Lgbm Randomoversampler
0.7584 ±0.0815
Lgbm Baseline
0.7664 ±0.0765
Xgboost Baseline
0.7618 ±0.0788
Xgboost Smote
0.7740 ±0.0709
Randomforest Baseline
0.7831 ±0.0587
Randomforest Smote
0.7885 ±0.0651
Extratrees Smote
0.8215 ±0.0617
Extratrees Baseline
0.8406 ±0.0890
0.0
0.2
0.4
0.6
0.8
1.0
F1-Score (Macro)


Model Stability Analysis
Lower = More Stable
High Stability
Moderate Stability
Extratrees Baseline
Xgboost Baseline
Xgboost Smote
Lgbm Randomoversampler
Randomforest Smote
Extratrees Smote
Lgbm Baseline
Randomforest Baseline
0.0928
0.0871
0.0798
0.0785
0.0779
0.0768
0.0735
0.0669
0.00
0.02
0.04
0.06
0.08
0.10
Standard Deviation
Performance vs Stability Trade-off
(10-Fold CV)
0.10
0.09
0.08
0.07
0.06
0.05
Standard Deviation
Extratrees Baseline
Xgboost Baseline
Xgboost Smote
Lgbm Randomoversampler
Randomforest Smote
Extratrees Smote
Lgbm Baseline
Randomforest Baseline
0.76
0.78
0.80
0.82
0.84
Mean Balanced Accuracy

Mix Dataset - Detailed Performance Metrics

Rank	Model	Bal Acc (Mean±Std)	F1 (Mean±Std)	Accuracy (Mean±Std)	Stability
1	Extratrees Baseline	0.8520±0.0928	0.8406±0.0890	0.8713±0.0579	Moderate
2	Extratrees Smote	0.8354±0.0768	0.8215±0.0617	0.8640±0.0411	Moderate
3	Randomforest Smote	0.8175±0.0779	0.7885±0.0651	0.8311±0.0495	Moderate
4	Randomforest Baseline	0.8025±0.0669	0.7831±0.0587	0.8288±0.0384	Moderate
5	Xgboost Smote	0.7885±0.0798	0.7740±0.0709	0.8288±0.0496	Moderate
6	Xgboost Baseline	0.7781±0.0871	0.7618±0.0788	0.8242±0.0494	Moderate
7	Lgbm Baseline	0.7640±0.0735	0.7664±0.0765	0.8405±0.0319	Moderate
8	Lgbm Randomoversampler	0.7514±0.0785	0.7584±0.0815	0.8311±0.0450	Moderate